AI VISION VIAD

Visually Impaired Assistive Device

Qassim University College of Computer | كليـــة الحـــاسب جامعة القصيم
Department of Computer Science

Rayan Altawijari 431107823| Faris Alsuhibani 431107834| Mohammed Alsharekh431108100
Muhiya Almutairi 431107802| Abdullah Alfayez 431108148
Supervisor: Dr. Ali Mustafa Qamar Khan

Qassim University Applied College الكلية التطبيقية جامعة القصيم
مركز فكرة

intruduction

The **AI VISION VIAD** is a wearable assistive device designed to provide independence for visually impaired users through real-time environmental awareness. Globally, over 295 million people suffer from visual impairment, and current tools often face issues. This project solves these challenges by using a **Raspberry Pi 4** to combine **SSD-MobileNet object detection** with ultrasonic sensors. This hybrid approach ensures critical safety alerts work offline, while a cloud-based AI assistant provides detailed scene descriptions when connected.

1.Abstract

AI VISION VIAD is an AI-powered smart glasses system designed to enhance environmental awareness for visually impaired users. By fusing computer vision and ultrasonic sensing, the device identifies obstacles and objects in real-time, providing .audio feedback via a hybrid voice interface

2. Problem Statement

With 295 million people globally facing visual impairment, existing tools have significant gaps:

- **High Cost**: Commercial smart glasses up to $2000
- **Blind Spots**: Canes don't detect head-level hazards
- **Offline Limits**: Most devices require internet

3. Project Scope & Aim

Build an affordable, hands-free wearable that:

✔ Detects obstacles using SSD-MobileNet V2
✔ Measures distances using ultrasonic sensors
✔ Runs offline for simple feedback
✔ Runs online for complex feedback
✔ Affordable, lightweight, wareable design

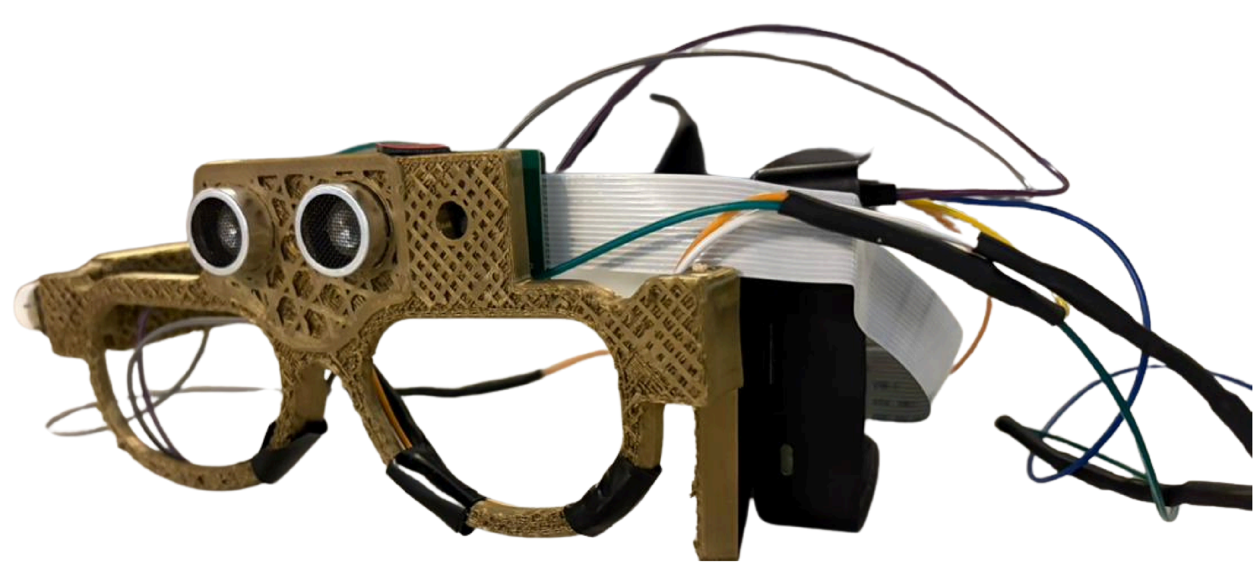
4. Hardware Components

Raspberry Pi 4
Main processor

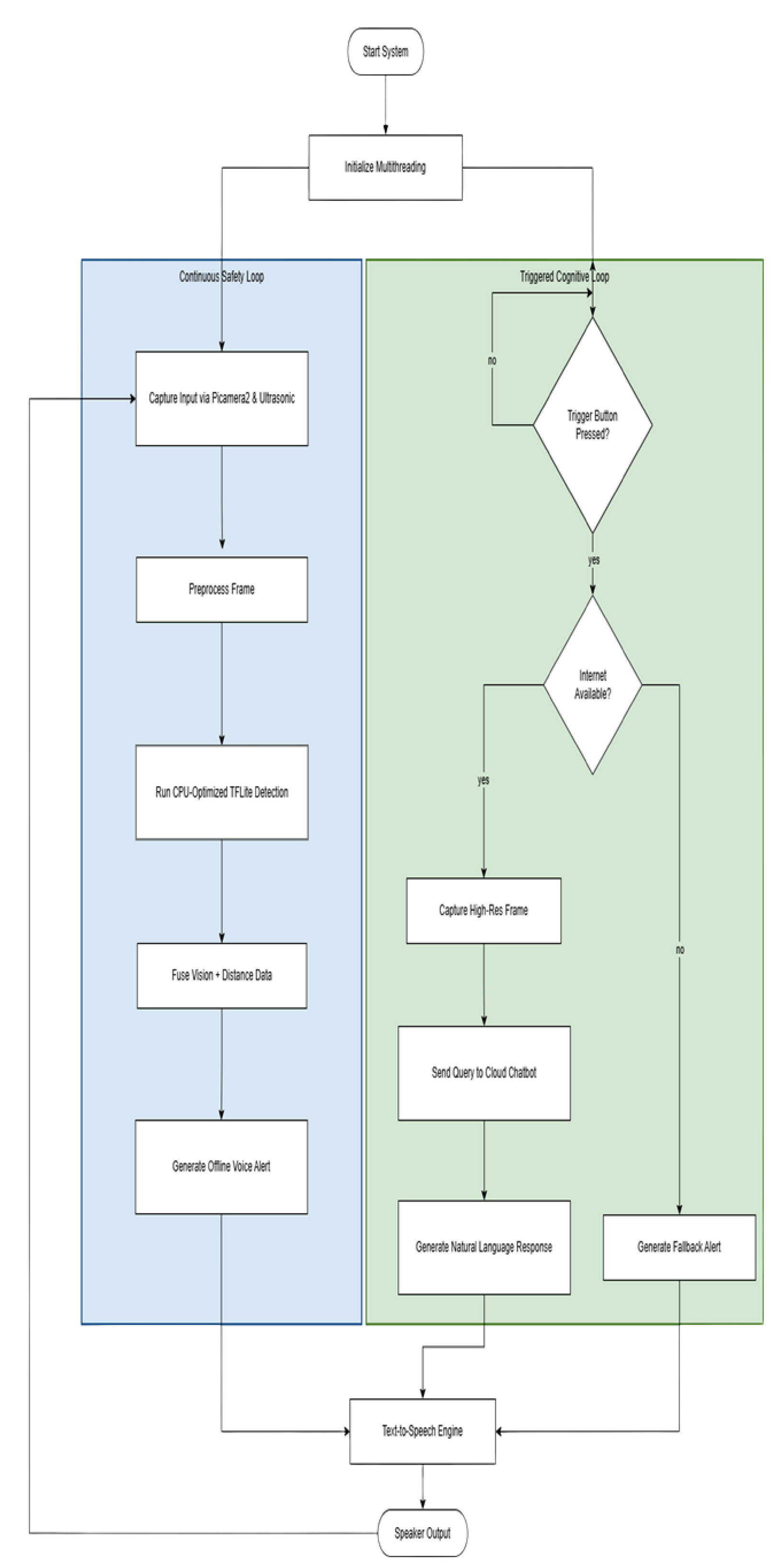
Pi Camera V2

HC-SR04
Ultrasonic sensor

bluetooth earbuds



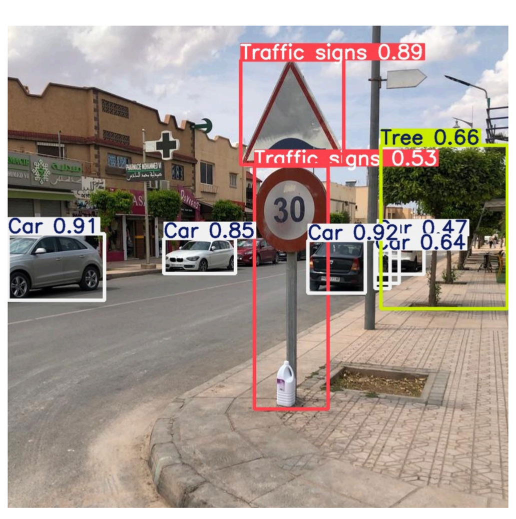
5. System Architecture

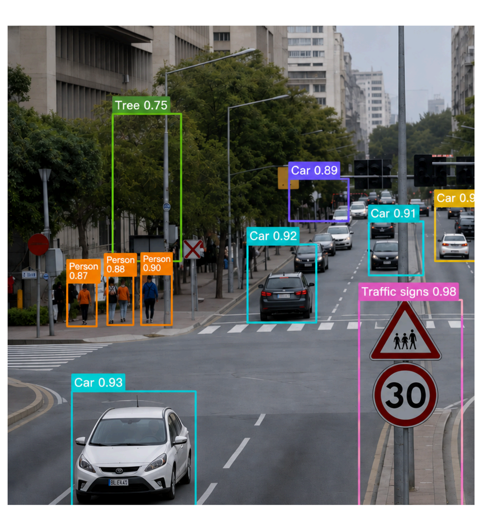


6. Detection Performance

Object Detection Accuracy 87%

Distance Accuracy (±5cm) 94%





7. Experimental Results

Offline System Latency	150 ms
Online System Latency	2.5 sec
Model Accuracy (SSD-MobileNet)	87%
CPU Temperature (Passive)	68°C

8. Key Features

- Hybrid Voice Interface (online & offline)
- Low Cost
- AI voice chatbot
- Hands-Free Glasses form factor

9. Future Work

- GPS navigation integration
- Built-in Small Language Model(SLM)
- Face recognition for social contexts
- Miniaturized hardware design

10. Tools & Technologies

OpenCV

3D PRINTING

COCO Common Objects in Context

Conclusion

The AI VISION VIAD successfully bridges the gap between reactive safety and cognitive understanding. Testing proved the system is highly reliable, with local detection achieving over 87% accuracy and a fast 150ms latency for immediate hazards. By fusing camera data with ultrasonic distance sensing, the device offers "head level" protection that traditional canes cannot provide. This project establishes a scalable, affordable foundation for future assistive tech, moving closer to a world where everyone can navigate their surroundings with confidence.